Cap 6 - Teoria Conjuntos

Exemplos

Ex 6.1 Apenas explicativo

Ex 6.2 Apenas explicativo

Ex 6.3 Apenas explicativo

Ex 6.4 Apenas explicativo

Ex 6.5 $A = \{2, 3, 5, 7, 11, 13, 17, 19\}$

Ex 6.6 a) $A = \{a, e, i, o, u\}$

b) $N = \{0, 1, 2, 3, \ldots\}$

c) Subjetivo

d) $\begin{cases} a = 1 \\ b = -3 \\ c = 2 \end{cases}$ $R = \{1, 2\}$

Ex 6.7 $A = \{x \in \mathbb{N} \mid 5 \leq x \leq 10\}$

Ex 6.8
$A = \{1, 3\}$
$B = \{1, 2, 3\}$
$C = \{1, 2, 3\}$

Relações de inclusão:

$$\begin{cases} A \subset B \\ A \subset C \\ B \not\subset A \\ B \subset C \\ C \not\subset A \\ C \subset B \end{cases}$$

Como $A \subset B$ e $A \subset C$, então A é subconjunto próprio de B e de C

Ex 6.9
$A = \{a, e, i, o, 3, 5\}$
$B = \{a, 3, 4\}$

$$A \cup B = \{a, e, i, o, u, 3, 4, 5\}$$

Ex 6.10 a) $A = \{a, e, i, o, 3, 5\}$
$B = \{a, 3, 4\}$

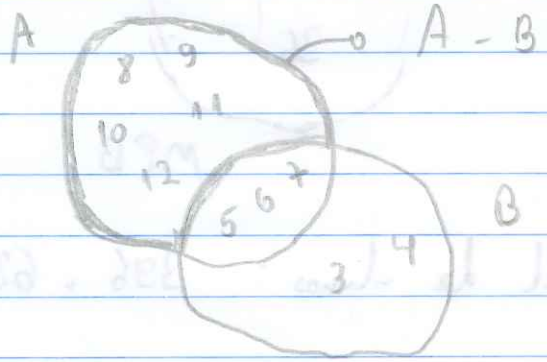
$$A \cap B = \{a, 3\}$$

b) $A = \{a, e, i, o, 3, 5\}$
$B = \{u, 4\}$

$$A \cap B = \emptyset$$

Ex 6.11

$A = \{5, 6, 7, 8, 9, 10, 11, 12\}$
$B = \{3, 4, 5, 6, 7\}$



$$A - B = \{8, 9, 10, 11, 12\}$$

Ex 6.12

$A = \{0, 1, 2, 3\}$
$B = \{2, 3\}$

a) $C_A^B = A - B = \{0, 1\}$

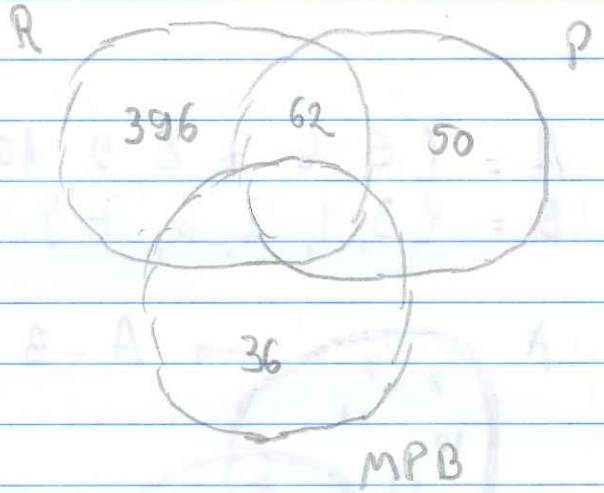
b) $C_B^A = B - A = \emptyset$

04

Ex 6.13

Rock = 458 alunos
Pop = 112 "
MPB = 36 "

Rock e pop = 62



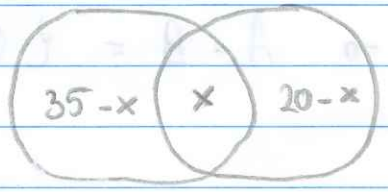
Total de alunos: 396 + 62 + 50 + 36

644 alunos entrevistados

Ex 6.14

Pinturas = 35
Esculturas = 20
Nenhum = 5

Total = 42



N = 5

$35 - x + x + 20 - x + 5 = 42$

$x = 18$

Pinturas (apenas): 17

Ex 6.15 Apenas explicativo

Ex 6.16 $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

a) $[\{1, 3, 5\}, \{2, 6\}, \{4, 8, 9\}]$

Não é partição de S, pois o elemento 7 não $\in$ a nenhum dos subconjuntos

b) $[\{1, 3, 5\}, \{2, 4, 6, 8\}, \{5, 7, 9\}]$

Não é partição de S porque o elemento 5 pertence a 2 subconjuntos

c) $[\{1, 3, 5\}, \{2, 4, 6, 8\}, \{7, 9\}]$

É partição de S.

Ex 6.17 $A = \{a, 1, 5\}$

$$A^2 = \{(a,a), (a,1), (a,5), (1,a), (1,1), (1,5), (5,a), (5,1), (5,5)\}$$

Lista 6 - Teoria dos Conjuntos

01) a) $\{1, 3, 5, \ldots\}$

b) $\{3, 4, 5, \ldots, 19\}$

c) $\{\ldots, -10, -5, 0, 5, 10, \ldots\}$

02) a) $\{x \in \mathbb{N} / x \leq 9\}$

b) $\{x \in \mathbb{Z} / x^2 = 1\}$

c) $\{x \in \mathbb{N} / 1 \leq x \leq 7 \text{ e } x \text{ é impar}\}$

d) $\{x \in \mathbb{N} / x \leq 6 \text{ e } x \text{ é par}\}$

e) $\{x \in \mathbb{N} / x = \alpha^2, \alpha \in \mathbb{N}\}$

f) $\{x \in \mathbb{N} / x = 7\alpha + 3, \alpha \in \mathbb{N}\}$

03) a) $\{-1, 0, 1, 2, 3, 4, 5, 6, 7\}$

b) $\{0, 2, 4, 6, 8\}$

c) $\{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$

d) $\emptyset$

e)

$$x^2 + 2x + 1 - (x^2 - 2x + 11) = 4x$$

$$4x = 4x$$

(Z)

04) a) F

b) V

c) F

d) F

e) V

f) F

g) V

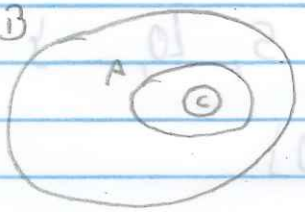
h) V

(02)

(05) a) $A \not\subset B$
b) $A \not\subset B$
c) $A \subset B$
d) $A \subset B$

(06) c) $\{0\}$

(07)



e) C é subconjunto de B

(08) a) $A = \{\ldots, -4, -2, 0, 2, 4\}$

b) $B = \{2, 3, 5, 7, 11, 13, \ldots\}$

c) $C = \{2\}$

d) $D = \{6, 8, 10\}$

(09)

$D = \{x \in \mathbb{N} / 2 < x < 3\} = \emptyset$

(10) $A = \{x \in \mathbb{N} / x^2 = 9\} \rightarrow x = 3 \rightarrow$ UNITÁRIO

$B = \{x \in \mathbb{Z} / x^2 + x = 2\} \rightarrow x' = 1,\ x'' = -2$ } NÃO UNITÁRIO

$C = \{x \in \mathbb{R} / x^2 = 5\} \rightarrow x = \pm\sqrt{5} \rightarrow$ NÃO UNITÁRIO

$D = \{x \in \mathbb{N} / 2 < x < 3\} = \emptyset$. NÃO UNITÁRIO

$E = \{x \in \mathbb{Z} / x = 0\}$. UNITÁRIO

(II) $A = \{0, 1, 2, 3, 4, 5, 6\}$

$B = \{0, 1, 2, 3, 4, 5, 6, 7\}$

$H =$ 

(i) $A \subset B$
$A \subset H$

(ii) $B \not\subset H$

(12) $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
$B = \{1, 3, 5, 7, 9\}$
$C = \{-1, -2, 1, 3, 5, 7, 9\}$

a) V
b) F
c) V
d) V
e) F
f) F
g) V
h) V
i) V
j) F
k) V

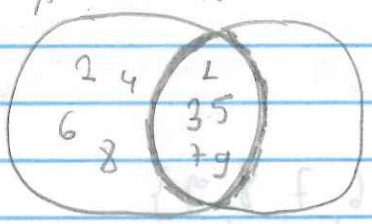
(13) a) $A \cap B = \{1, 3, 5, 7, 9\}$
b) $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
c) $C \cap B = \{1, 3, 5, 7, 9\}$
d) $A \cap C = \{1, 3, 5, 7, 9\}$
e) $C \cup B = \{-1, -2, 1, 3, 5, 7, 9\}$
f) $A \cup C = \{-1, -2, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
g) $A - B = \{2, 4, 6, 8\}$
h) $C - B = \{-1, -2\}$
i) $B - A = \{\}$
j) $A - (B \cap C) = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} - \{1, 3, 5, 7, 9\}$

$\{2, 4, 6, 8\}$

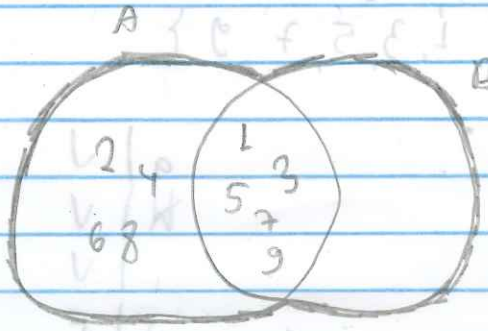
05

Representação da Questão (12) através de diagramas de Venn

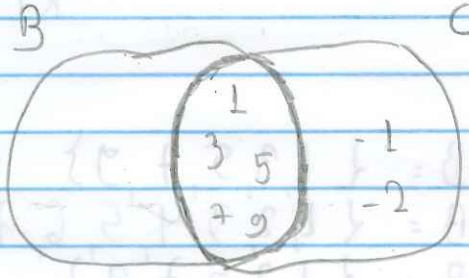
a) $A \cap B$



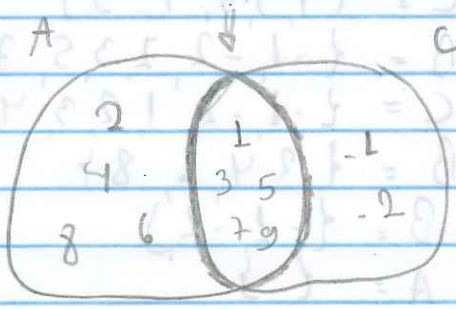
b) $A \cup B$



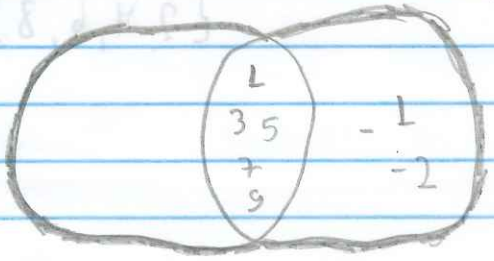
c) $C \cap B$



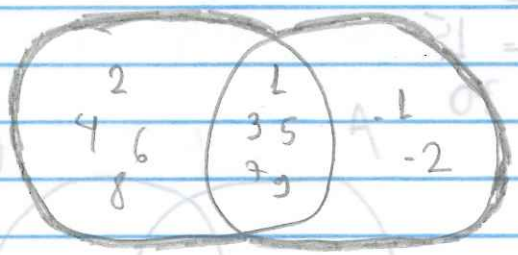
d) $A \cap C$



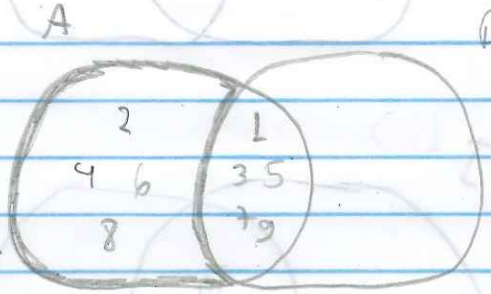
e) $C \cup B$



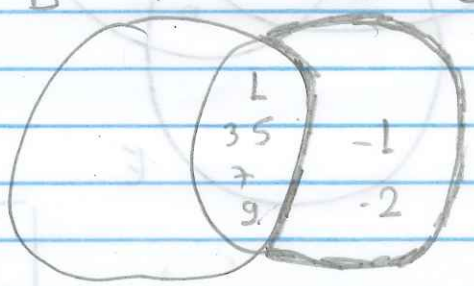
f) $A \cup C$



g) $A - B$



h) $C - B$



(14) $V = 300$
$P = 180$
$C = 70$

$C_v(P \cup C) \Rightarrow 300 - 180 - 70$

$C = 50$

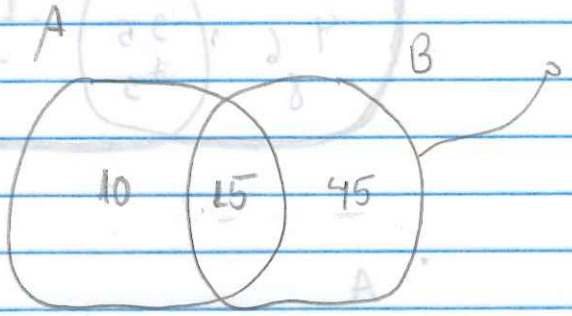
Resp: 50 pessoas c/ cabelos loiros

07

15) $A = 25$

$A \cap B = 15$

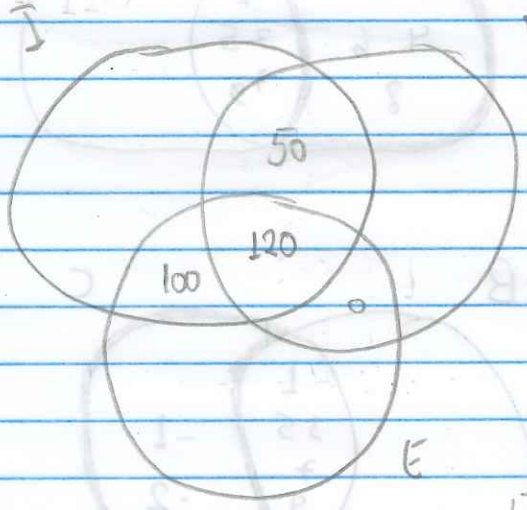
$A \cup B = 70$



Total = 70

B = 60 elementos

16)

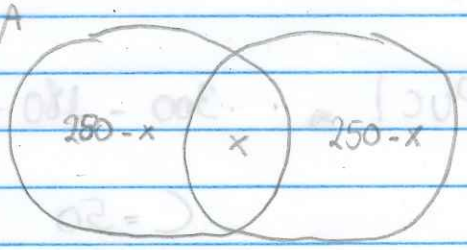


U = 1500

1500
100
120
50
1230

Resp: 1230 candidatos falam somente 1 língua

17)



V = 500

$280 - x + x + 250 - x = 430$

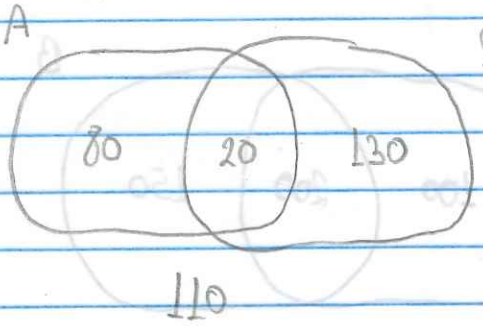
$530 - 430 = x$

$x = 100$

$A = 280 - 100 = 180$

Resp: 180 pessoas assistem exclusivamente o canal A

18



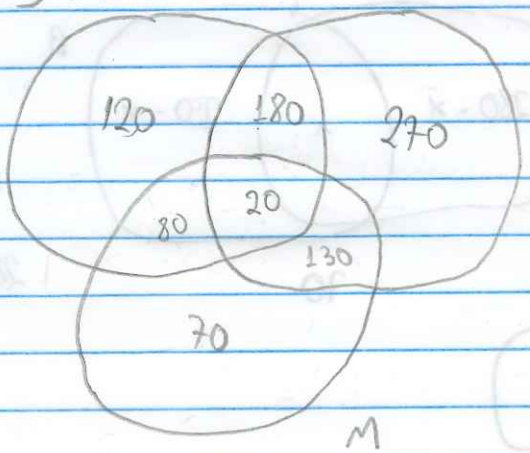
V = ??

80 + 20 + 130 + 110

V = 340

Resp: Total = 340 pessoas

19



V = 1000

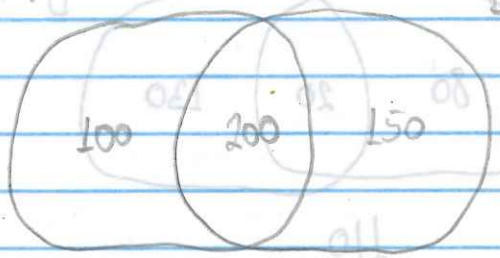
a) 120 + 70 + 270 = 460 pessoas

b) 1000 − (120 + 80 + 20 + 180 + 130 + 70 + 270) = 130 pessoas

c) 80 + 20 + 180 + 130 = 410 pessoas

09

20

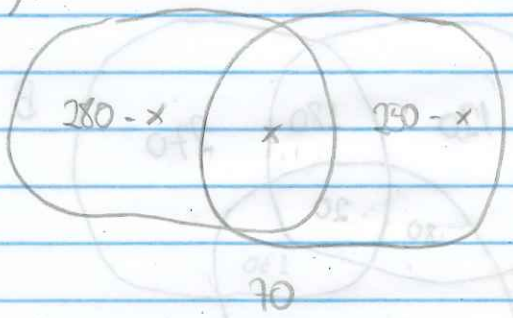


$V = 500$

$500 - 100 - 200 - 150 = 50$

50 clientes

21



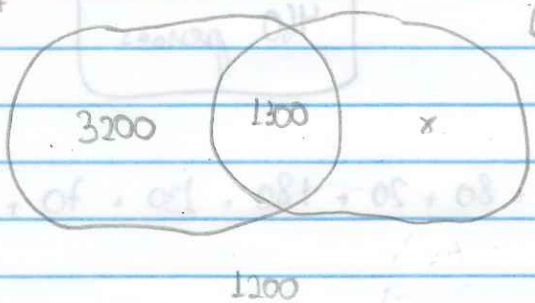
$V = 500$

$280 - x + x + 250 - x = 430$

$x = 100$

100 pessoas

22



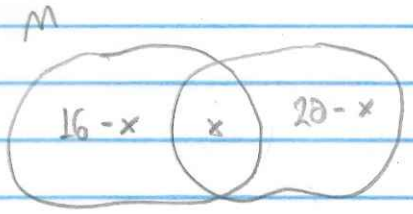
$V = 10000$

$3200 + 1300 + x = 8800$

$x = 4300$

a) 3200 pessoas
b) 5600 "
c) 4300 "

(23)

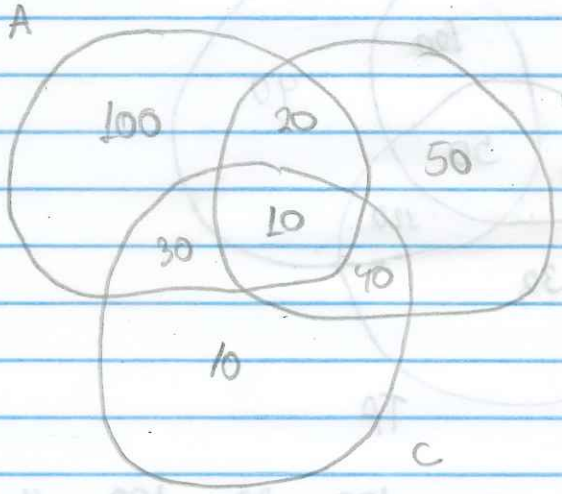


$V = 30$

$16 - x + x + 20 - x = 30$

$36 - 30 = x \Rightarrow$ **$x = 6$ alunos**

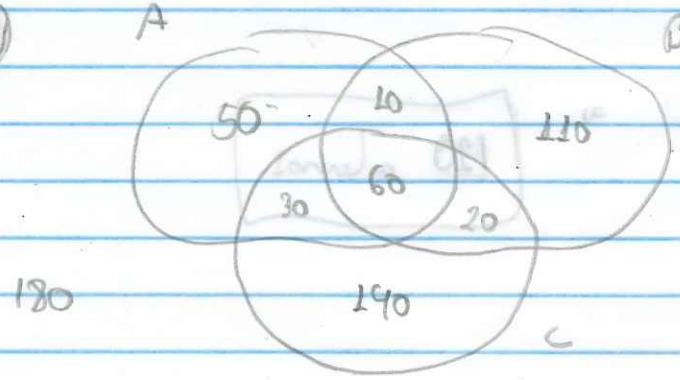
(24)



$V = 300$

$300 - (100 + 20 + 10 + 30 + 30 + 40 + 10) =$ **40 func. Letra d**

(25)



$U = 600$

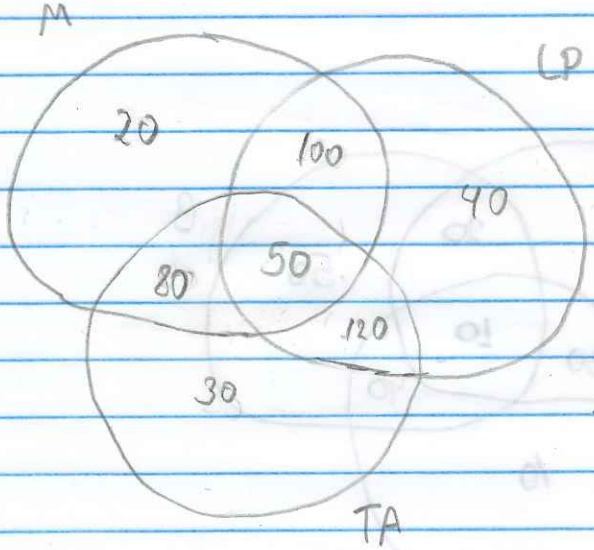
$U = 50 + 30 + 10 + 60 + 20 + 110 + 140 + 180 \Rightarrow$ **600** pessoas

(11)

a) 50 pessoas
b) 420 "
c) 280 "
d) 140 "
e) 600 "

(26)

U = 700 alunos



a) 700 − (20 + 80 + 50 + 100 + 30 + 120 + 40) = 260 alunos

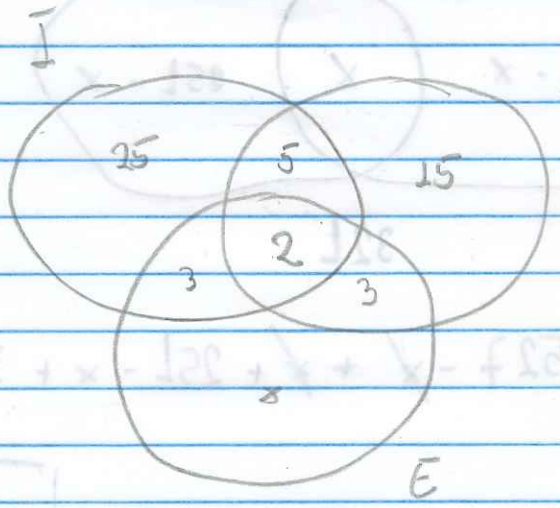
b) 20 + 80 + 50 + 100 + 120 + 30 = 400 alunos

c) = 20 + 100 ⇒ 120 alunos

27) a) F
b) V
c) F
d) V

28) a) F
b) F
c) V
d) F
e) F

29)



V = 80 alunos

7 = N

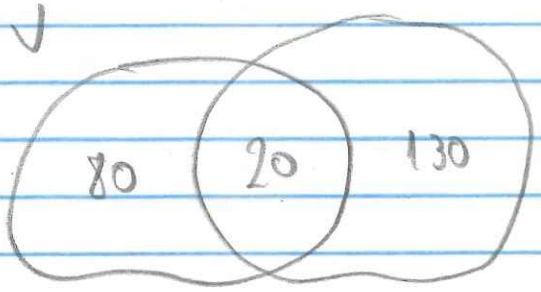
$25 + 5 + 2 + 3 + 3 + 15 + x = 73$

$x = 20$ cursam somente Espanhol

a) F
b) V
c) V
d) V

(13)

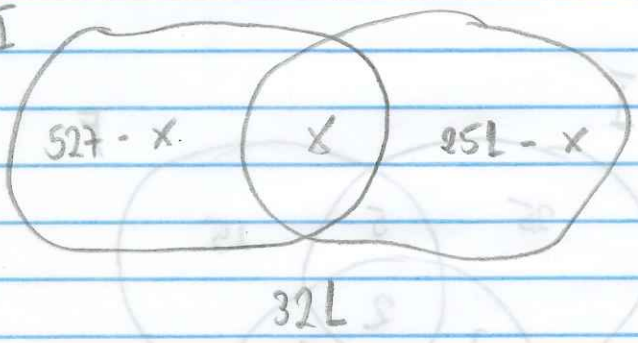
(30)



110 = Nenhum

$130 + 80 + 20 + 110 =$ **340 alunos**

(31)

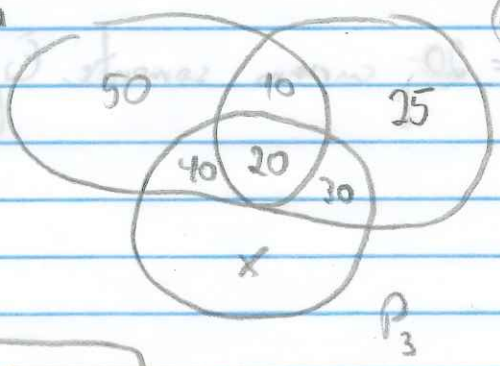


V = 979

$527 - x + x + 251 - x + 32L = 979$

x = 120 candidatos

(32)



V = 200

a	25	candidatos
b	100	"
c	85	"

$50 + 40 + 20 + 10 + 30 + 25 + x = 200$

$x = 25$

(33) $(x-2) \cdot \frac{3}{2} \rightarrow \frac{3x-6}{2} + 1 \rightarrow \frac{3x-6+2}{2} \rightarrow \frac{3x-4}{2}$

$\frac{3x-4}{x} \cdot x \rightarrow 3x-4-1 \rightarrow (3x-5) \cdot 3 = 21$

$9x - 15 = 21$

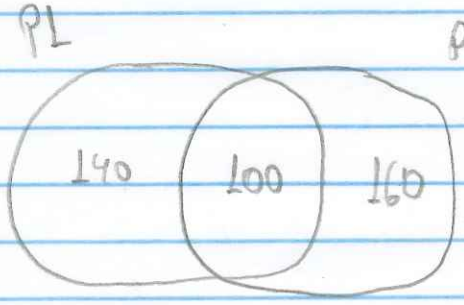
$9x = 36$

$x = 4$

$4 \in \{1, 2, 3, 4\}$

Letra A

(34)



Nenhum = 50

Total = 140 + 160 + 100 + 50

Total = 450 alunos